

Value at Risk & GARCH Volatility

Folder: VaR_analysis | Static vs time-varying volatility VaR back-test

1. Objective

Show why a static-volatility Value at Risk (VaR) fails when volatility changes through time, and how a GARCH(1,1) model corrects it. The folder holds VaR_GARCH.ipynb and the comparison animation VaR_GARCH.gif.

2. Synthetic data

- 2000 daily prices from geometric Brownian motion, $S_{t+1} = S_t \exp((\mu - \sigma^2/2) dt + \sigma \sqrt{dt} X)$, drift $\mu = 0.0004$.
- The volatility path is deliberately regime-switching: 0.8% baseline, a 3.0% crisis over days 600-900 and a 2.2% aftershock over days 1400-1600.
- Log returns $r_t = \ln(S_t/S_{t-1})$ are Gaussian with a mean and variance set by the volatility at that time.

3. Method

- **Static VaR:** fit a single normal to all returns and take the 1% quantile (cross-checked against the empirical sorted-return quantile). A conditional VaR (expected shortfall) is defined as the mean return below that quantile.
- **GARCH(1,1):** $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ is fitted to the returns. The notebook reports $\omega = 1.68e-6$, $\alpha = 0.0875$, $\beta = 0.9022$, giving persistence $\alpha + \beta = 0.990$ and a long-run sigma of about 1.28%.
- **Dynamic VaR:** each day's 1% VaR is the normal quantile using that day's GARCH σ_t and mean.

4. Back-test

Breaches (days on which the realised return falls below the VaR) are counted for each method. At a 1% level on about 2000 days roughly 20 breaches are expected. The static VaR breaches far more often during the crisis windows, because one global sigma understates risk when volatility spikes, while the GARCH VaR tracks the volatility and gives a breach count near the nominal rate.

5. Visualisation

Each frame of the animation has three panels: the return histogram with static (red) and GARCH (green) normal fits and their VaR lines, the return series with the static VaR line and a running breach count, and the return series against the moving GARCH VaR. It is rendered with Pillow every eighth day.

6. Files

File	Description
VaR_GARCH.ipynb	Data generation, static and GARCH VaR, breach counts, animation code
VaR_GARCH.gif	Animated comparison of static and GARCH VaR